

Title: To design a 8x1 multiplexer, use 2x1 multiplexors to design an 8x1 multiplexor and to create an 8 bit encoder in Logisim

Theory:

A multiplexer is a combinational circuit that has many data inputs and a single output, depending on control or select inputs. For N input lines, $\log_2(N)$ selection lines are required, or equivalently, for 2^n input lines, n selection lines are needed.

- Multiplexers are also known as "N-to-1 selectors," parallel-to-serial converters, many-to-one circuits, and universal logic circuits.

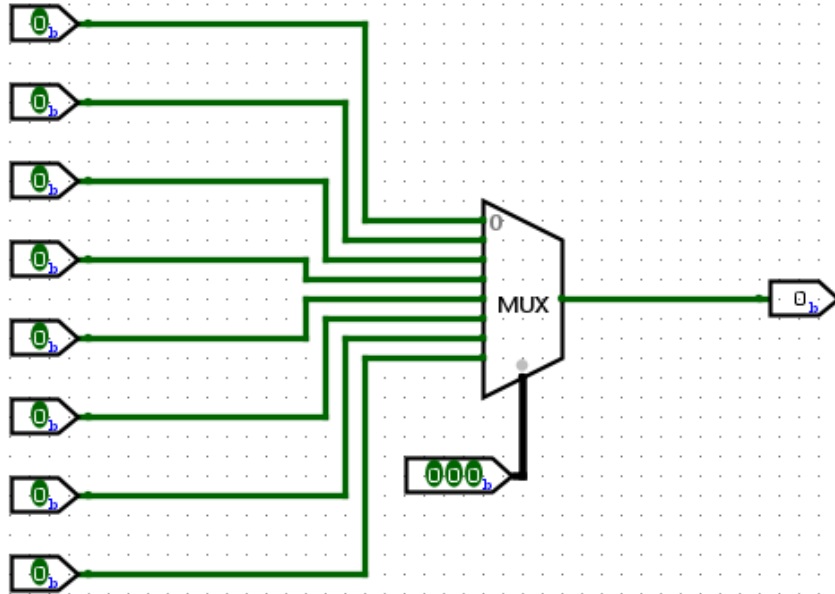
Due to not having a specific size MUX can come in various sizes for its inputs and larger MUXs can be made through Smaller MUXs

An encoder is a digital combinational circuit that converts multiple input signals into a binary code. It typically has one active input at a time and generates a binary output representing the position of that active input. For example, a 4-to-2 encoder has four inputs and produces a 2-bit output code. Encoders are useful in digital systems for data compression and efficient transmission. Variants like priority encoders assign importance to inputs to manage simultaneous signals.

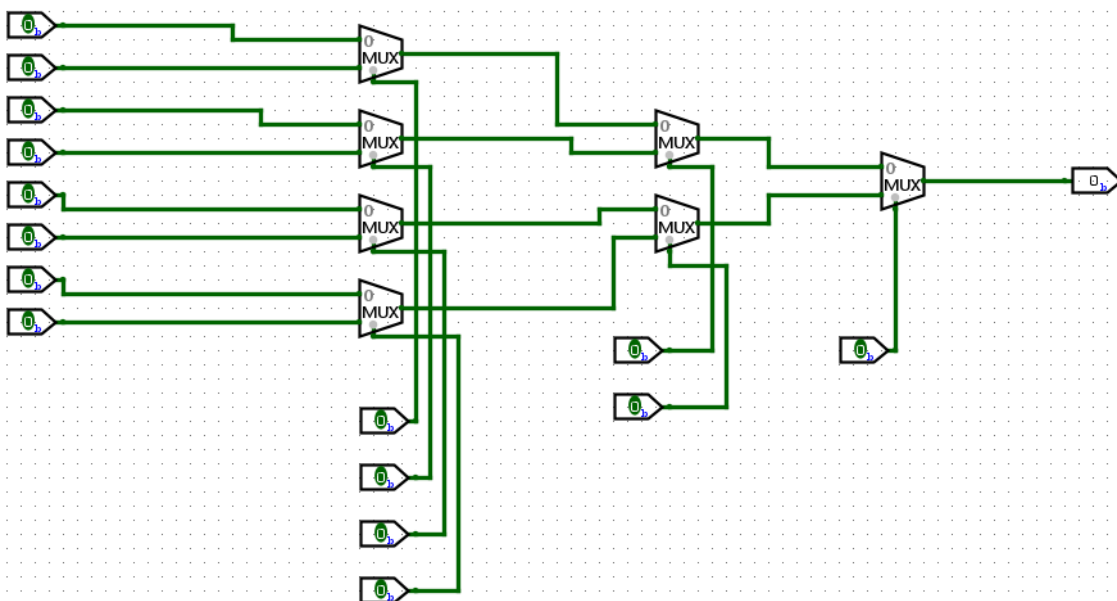
- The number of inputs is usually 2^n , with n output lines.
- Inputs are usually active high, meaning high voltage represents an active signal.

Similarly to a MUX an encoder come in various sizes

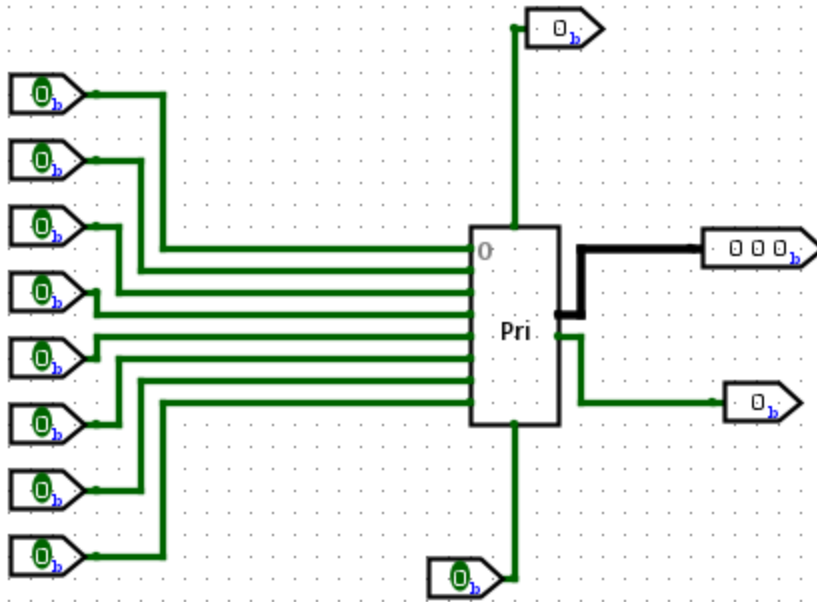
Implementation:



8 bit input 3 bit select line , 1 bit output



Building 8x1 Mux using 2X1 Mux requires 7 2x1 MUX,



South Input is an enable line , North Output shows if encoder is enabled to the user , 3 bit output east, 1 bit east output shows if main output is valid or not (at least 1 input line is active)

Results:

Successfully created Mux and Encoder in Logisim